

Department of Electrical and Electronics Engineering

University of Dhaka
Batch - 103, EEE-58

2nd Year 1st Semester



Course: EEE-2103

Title: Digital Electronics

Class Teacher: Md. Shahin Parvej

Note by
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Session: 2023-2024
Class Roll: SH-042

SYLLABUS

Introductory Concepts: Digital and Analog Quantities, Binary Digits, Logic Levels, Digital Electronic Signals and Switches, Basic Logic Functions, Combinational and Sequential Logic Functions, Introduction to Programmable Logic and Fixed-Function Logic Devices.

Number Systems, Operations, and Codes: Decimal, Binary, Hexadecimal, Octal and Binary Coded Decimal Number Systems, Binary Arithmetic and Digital Codes.

Logic Gates: Inverter, AND Gate, OR Gate, NAND Gate, NOR Gate, Exclusive-OR and Exclusive-NOR Gate, Universal Gate, Programmable Logic, Fixed-Function Logic Gates.

Boolean Algebra and Logic Simplification: Boolean Operations, Expressions, Laws and Algebra, De Morgan's Theorems, Boolean Analysis of Logic Circuits, The Karnaugh Map, Don't Care Conditions, SOP and POS Minimization.

Arithmetic Operations and Circuits: Half and Full Adders, Half and Full Subtractor, 2's complement, Ripple Carry and Look-Ahead Carry Adders, BCD Adders, Cascading BCD Adders, Multipliers.

Functions of Combinational Logic: Decoders: 1-of-8 decoders, BCD-to-Decimal Decoder/Driver, BCD-to-7-Segment Decoder/Drivers, Decoder IC, Encoders and Applications, Priority Encoders, Code Converters, Magnitude Comparators, Parity Generators/Checkers, Multiplexers: Two, Four, Eight, Sixteen, Quad Two-Input MUX, Multiplexer Applications: Data Routing, Parallel-to-Serial Conversion, Logic Functions Generation, Demultiplexers and its Applications, 1-Line-to-8-Line Demultiplexer, Synchronous Data Transmission.

Sequential Logic: NAND/NOR Gate Latch, Edge-Detector Circuit, Flip-Flops: Clocked S-C Flip-Flop, Clocked J-K Flip-Flop, Clocked

D Flip-Flop, T Flip-Flop, Master/Slave Flip-Flop, Clocked J-K Flip-Flop with Asynchronous Inputs, FlipFlop Applications: Edge Detection, Switch Bouncing Reduction, Parallel Data Transfer, Serial Data Transfer. Shift Registers and Counters: Shift Register Operations, Types of Shift Register, Shift Register Counters: Ring Counter and Johnson Counter, Asynchronous Counters, IC Asynchronous Counters, Asynchronous Down Counters, Synchronous Up/Down Counters, Pre-settable Counters, Counter with Arbitrary Sequences, Design of Synchronous Counters and Cascaded Counters.

Interfacing with the Analog World: Methods of Digital-to-Analog Conversion: Weighted Resistors Converter, R-2R Ladder Converter, Methods of Analog-to-Digital Conversion: Flash, Digital Ramp, Successive Approximation Converter, ADC and DAC Applications.

Integrated-Circuit Logic Families: Characteristic Parameters, The TTL Logic Family, TTL Series Characteristics, TTL Circuit Operation, Current-Sinking and Sourcing Action, Totem-Pole Output Circuit, TTL Series Characteristics, TTL Loading, Fan-In and Fan-Out, TTL NOR Gate, TTL NAND Gate, Digital MOSFET Circuits: CMOS, NMOS and PMOS Gates, Emitter-Coupled Logic, Direct Coupled Transistor Logic, Resistor Transistor Logic.

Finite State Machines: State Diagram, Moore-type and Melay-type Machines, State Machines Synthesis, State Machines in Verilog, State Encoding.

Introduction to Programmable Logic Device: Programmable Logic Device (PLD), Programmable Logic Array (PLA), Field-Programmable Gate Array (FPGA).

REFERENCE BOOKS

1. Digital Systems, 12th Edition, Ronald Tocci, Neal Widmer and Greg Moss, Pearson.
2. Digital Fundamentals, 11th Edition, Thomas L. Floyd, Pearson.
3. Digital Electronics: A Practical Approach, 8th Edition, William Kleitz, Prentice Hall.
4. Digital Electronics: Principles, Devices and Applications, Anil K. Maini, John Wiley and Sons Pub. Ltd.

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CHAPTER 1

INTRODUCTORY CONCEPTS